Marta Cipriani

Curriculum Vitae et Studiorum

Academic experience

2026–Present **Post-Doctoral Research Fellow**, *School of Mathematics and Statistics, University College Dublin*.
Project: "Innovative statistical and chemometric methods for high-dimensional, agri-food data."
Principal Investigator: Prof. Claire Gormley

2024–2025 **Research Fellow in Statistics**, *Dpt. of Political Sciences, Roma Tre University*.
Project: "How will the circular economy transition affect the dynamics of the labour market: the role of immigration."
Supervisor: Prof. Francesco Lagona

2024 **Visiting Graduate Researcher**, *Mathematical Institute, Leiden University*.
Project: "Multiple imputation in cure models with application to childhood osteosarcoma."
Supervisor: Prof. Marta Fiocco

Education

2022–2026 **PhD in Methodological Statistics with mention of "Doctor Europaeus"**, *Sapienza University of Rome*, Thesis title: *"Flexible methods for survival data: synthetic patients, imputation and prediction in mixture cure models"*. Final grade: Cum laude.
Supervisor: Prof. Marco Alfò

2023 **Course "Pseudo-observations in survival analysis"**, *University of Milano-Bicocca*.
Instructors: Per Kragh Andersen, Henrik Ravn

2019–2022 **Master's degree in Statistical Sciences - Major: Biostatistics**, *Sapienza University of Rome*, Thesis title: *"Joint models and pseudo-observations in survival analysis"*. Final grade: 110/110 cum laude.
Supervisor: Prof. Marco Alfò

2018–2019 **Erasmus+ Program**, *University Paris I Panthéon-Sorbonne*, .

2016–2019 **Bachelor's degree in Political Sciences**, *Roma Tre University*, Thesis title: *"The Cox model applied to a study on health and longevity in China"*. Final grade: 110/110 cum laude.
Supervisor: Prof. Francesco Lagona

Work experience

2022–Present **Biostatistical consultant**, *Fondazione GIMEMA Franco Mandelli ONLUS*, Rome.
Support in statistical methodology for clinical trials conducted by the Foundation.

Teaching experience

Teaching Assistant

2024–2025 **Statistica**, *Dpt. of Political Sciences, Roma Tre University*.
Course for the Bachelor programmes in L-36 and L-37

2023–2024 **Statistica**, *Dpt. of Political Sciences, Roma Tre University*.
Course for the Bachelor programmes in L-36 and L-37

2023–2024 **Statistica di base**, *Dpt. of Statistical Sciences, Sapienza University of Rome*.
Course for the Bachelor program in L-41

Funded research projects and Scholarships

Principal investigator

2025 **Research Grant for Junior Researcher**, *Sapienza University of Rome*.
Title: Dynamic prediction in mixture cure models: a model-based landmarking approach

2024 **Research Grant for Junior Researcher**, *Sapienza University of Rome*.
Title: A multiple imputation approach to distinguish curative from life-prolonging effects in the presence of missing covariates

2023 **Research Grant for Junior Researcher**, *Sapienza University of Rome*.
Title: Unlocking the potential of artificial intelligence for generating synthetic patients in clinical trials

2022 **Full three-year PhD scholarship**, *Sapienza University of Rome*.

Research collaborator

2024 **Major Research Projects**, *Sapienza University of Rome*.
Title: Latent variable models for complex health data

2023 **Major Research Projects**, *Sapienza University of Rome*.
Title: Advancing precision public health medicine: high-performance computing for efficient solutions to bioinformatics and biostatistical challenges

Awards

2025 **Best Poster Award**, *International Workshop on Statistical Modelling*, Limerick, Ireland.

2025 **Best Poster Award**, *Survival Analysis for Junior Researchers Conference*, Bonn, Germany.

2024 **Best Oral Paper Presentation**, *International Biometric Conference*, Atlanta, USA.

Journal referee

Biometrical Journal, Journal of Statistical Computation and Simulation (JSCS), Metron, Italian Statistical Society Series on Advances in Statistics, Journal of Dairy Science.

Personal information

Programming R, RStudio, Rmarkdown, LaTeX, Overleaf, SAS, Stata, HTML, Github, Python

Languages Italian: mother tongue; English: advanced; French: baccalauréat général; Spanish: intermediate

Memberships Società Italiana di Statistica (SIS), Young SIS (y-SIS), International Biometric Society (IBS), International Society for Clinical Biostatistics (ISCB)

Metrics Tot. citations: 95 (Scholar); H-index: 6 (Scholar) accessed on 10/02/2026

Conferences and Workshops

Invited talk

2025 **19th International Joint Conference CFE-CMStatistics 2025**, *London, United Kingdom*, Birkbeck, University of London.
Title: Extending landmarking to mixture cure models with longitudinal covariates

2025 **New Frontiers in Hematologic Research: Quality of Life and Artificial Intelligence - GIMEMA International Meeting**, *Sala della Protomoteca, Campidoglio*, Rome, Italy.
Title: Unlocking the potential of synthetic patients for accelerating hematologic research

2024 **Workshop on "Dati sintetici per accelerare la ricerca clinica"**, *Sala Convegni AIL*, Rome, Italy.
Title: Un esempio della ricerca GIMEMA

Invited seminar

2026 **Flexible methods for survival data**, *School of Mathematics and Statistics, University College Dublin*, Ireland.

2025 **Synthetic Data: Opportunities and Challanges**, *Dpt. of Political Sciences, Roma Tre University*, Italy.

2024 **A multiple imputation approach to distinguish curative from life-prolonging effects in the presence of missing covariates**, *Mathematical Institute, Leiden University*, The Netherlands.

Contributed

2025 **39th International Workshop on Statistical Modelling - IWSM 2025**, *University of Limerick*, Ireland.
Title: Extending landmarking to mixture cure model with time-varying covariates

2025 **SIS 2025 - Statistics for innovation**, *University of Genoa*, Italy.
Title: Missing covariates in mixture cure models: a multiple imputation approach

2025 **Survival Analysis for Junior Researchers Conference - SAfJR 2025**, *University of Bonn*, Germany.
Title: Extending landmarking to mixture cure model with time-varying covariates

2024 **International Biometric Conference - IBC 2024**, *Atlanta*, USA.
Title: A multiple imputation approach to distinguish curative from life-prolonging effects in the presence of missing covariates

2024 **Computational Intelligence methods for Bioinformatics and Biostatistics - CIBB 2024**, *University of Sannio*, Italy.
Title: Generating electronic health records using adversarial random forests

2024 **SIS 2024 – 52nd Scientific Meeting of the Italian Statistical Society**, *Università degli Studi di Bari "A. Moro"*, Italy.
Title: A flexible parametric approach to synthetic patients generation in clinical trials

2023 **SIS 2023 – Statistical Learning, Sustainability and Impact Evaluation**, *Università Politecnica delle Marche*, Italy.
Title: Pseudo-observations in survival analysis

Conference chair

2026 **Computational Intelligence methods for Bioinformatics and Biostatistics (CIBB) 2026**, Sapienza University of Rome

Organizing committee

2025 **GRASPA 2025**, Roma Tre University

2023 **Italian Bayesian Day**, Sapienza University of Rome

Programme committee

2025 **Computational Intelligence methods for Bioinformatics and Biostatistics (CIBB) 2025**, Politecnico di Milano

Session chair

2024 **International Biometric Conference - IBC 2024**, Session: Incomplete Data

2025 **Survival Analysis for Junior Researchers Conference - SAfJR 2025**, Session: Cure models

Publications

Scientific papers

2026 **Validation of ICC hierarchical classification in secondary AML**, *E. Attardi, M. Cipriani, L. Guarnera* et al.
Blood Advances. DOI: https://doi.org/10.1182/bloodadvances.2025018188

2025 **A flexible parametric approach to synthetic patients generation using health data**, *M. Cipriani, L. Di Rocco, M. Puopolo, M. Alfò*.
Statistical Methods and Applications, vol. 34, pp. 639–662. DOI: https://doi.org/10.1007/s10260-025-00800-5

2024 **Health-Related Quality of Life and Financial Burden in Ethiopian Patients With Chronic Myeloid Leukemia Receiving Tyrosine Kinase Inhibitors: A Cross-Sectional Study**, *F. Tadesse, F. Sparano, A. Gebremedhin, A. Abubeker, A. Piciocchi, M. Cipriani* et al.
JCO Global Oncology, 10:e2400281. DOI: https://doi.org/10.1200/GO-24-00281

2024 **Current Knowledge on Breast Implant-Associated Anaplastic Large Cell Lymphoma: Evidence from Italian Ministry of Health Registry Data**, *A. Campanale, M. Ventimiglia, M. Alfò, M. Cipriani, D. Minella, L. Lispi, A. Iachino* .
Aesthetic Plastic Surgery, vol. 49(5), pp. 1304–1313. DOI: https://doi.org/10.1007/s00266-024-04426-y

2024 **Health-related quality of life and symptoms of chronic myeloid leukemia patients after discontinuation of tyrosine kinase inhibitors: results from the EURO-SKKI Trial**, *F. Efficace, F.-X. Mahon, J. Richter, A. Piciocchi, M. Cipriani* et al.
Leukemia, vol. 38(8), pp. 1722–1730. DOI: https://doi.org/10.1038/s41375-024-02341-4

2024 **Unlocking the potential of synthetic patients for accelerating clinical trials: Results of the first GIMEMA experience on acute myeloid leukemia patients**, *A. Piciocchi, M. Cipriani, M. Messina, G. Marconi, V. Arena* et al.
eJHaem, vol. 5 (2), 353-359, 2023. DOI: https://doi.org/10.1002/jha2.873

2023 **A randomized double-blind clinical trial on safety and efficacy of tauroursodeoxycholic acid (TUDCA) as add-on treatment in patients affected by amyotrophic lateral sclerosis (ALS): the statistical analysis plan of TUDCA-ALS trial**, *F. L. Lombardo, S. Spila Alegiani, F. Mayer, M. Cipriani* et al.
Trials, vol. 24, article 792. DOI: https://doi.org/10.1186/s13063-023-07638-w

2023 **The Promotoer, a brain-computer interface-assisted intervention to promote upper limb functional motor recovery after stroke: a statistical analysis plan for a randomized controlled trial**, *M. Cipriani, F. Pichiorri, E. Colamarino, J. Toppi, F. Tamburella, M. Lorusso* et al.
Trials, vol. 24, article 736. DOI: https://doi.org/10.1186/s13063-023-07773-4

2023 **Applicability of 2022 classifications of acute myeloid leukemia in the real-world setting**, *E. Attardi, A. Savi, B. Borsellino, A. Piciocchi, M. Cipriani* et al.
Blood Adv, 7 (17): 5122–5131. DOI: https://doi.org/10.1182/bloodadvances.2023010173

Proceedings

2025 **Missing Covariates in Mixture Cure Models: A Multiple Imputation Approach**, *M. Cipriani, M. Fiocco, M. Alfó, E. Musta*.
Statistics for Innovation II, pp. 459–464, Springer, 2025. DOI: https://doi.org/10.1007/978-3-031-96303-2_75

2025 **A Flexible Parametric Approach to Synthetic Patients Generation in Clinical Trials**, *M. Cipriani, L. Di Rocco, M. Alfò*.
Methodological and Applied Statistics and Demography III, pp. 419–425, Springer. DOI: https://doi.org/10.1007/978-3-031-64431-3_69

2024 **Digital Twins As Control Groups to Accelerate Assessment of Safety and Efficacy of Novel Treatment Strategies in Ph-Negative Acute Lymphoblastic Leukemia**, *A. Piciocchi, M. Messina, G. Marsili, M. Cipriani* et al.
Blood, vol. 144 Suppl.1, pp. 274–275. DOI: https://doi.org/10.1182/blood-2024-200643

2023 **A Propensity Score Weighting Analysis Emphasizes the Progress of the Gimema Approach in Adult Ph-Positive Acute Lymphoblastic Leukemia in the TKI Era**, *A. Piciocchi, S. Chiaretti, M. Cipriani, M. Messina* et al.
Blood, vol. 142 Suppl.1, p. 4200. DOI: https://doi.org/10.1182/blood-2023-179102

2023 **Unraveling the Impact of 2022 Classifications on Secondary Acute Myeloid Leukemia: Assessing the True Qualification Power of Diagnostic Qualifiers**, *E. Attardi, A. Savi, B. Borsellino, M. Cipriani, A. Piciocchi* et al.
Blood, vol. 142 Suppl.1, p. 819. DOI: https://doi.org/10.1182/blood-2023-185709

2023 **P499: Applicability of 2022 classifications of acute myeloid leukemia in the real-world setting**, *E. Attardi, A. Savi, B. Borsellino, A. Piciocchi, M. Cipriani* et al.
HemaSphere, 7 (Suppl):e5402602. DOI: https://doi.org/10.1097/01.HS9.0000968904.54026.02

2023 **S159: Health-related quality of life of patients with chronic myeloid leukemia after discontinuation of tyrosine kinase inhibitors: Results from the EURO-SKI study**, *F. Efficace, F.-X. Mahon, J. Richter, A. Piciocchi, M. Cipriani* et al.
HemaSphere 7(S3):p e0972454. DOI: https://doi.org/10.1097/01.HS9.0000967548.09724.54

2023 **P523: The comparisong of VFLAI, FLAI and 3+7 regimens by multilevel propensity score weighting highlights the benefit of the addition of venetoclax in no low-risk AML treated in GIMEMA trials and real world**, *A. Piciocchi, G. Marconi, M. Cipriani* et al.
HemaSphere 7(S3):p e0034216. DOI: https://doi.org/10.1097/01.HS9.0000969000.00342.16

2023 **P495: Unlocking the potential of synthetic patients for accelerating clinical trials: Results of the first GIMEMA experience**, *A. Piciocchi, M. Cipriani, M. Messina, G. Marconi* et al.
HemaSphere 7(S3):p e4536536. DOI: https://doi.org/10.1097/01.HS9.0000968888.45365.36

2023 **Addition of venetoclax to FLAI induction chemotherapy (VFLAI) improves survival in patients with intermediate and high-risk AML vs FLAI and 3+7: A GIMEMA Italian cooperative analysis**, *G. Martinelli, M. Cipriani, G. Marconi, F. Guolo, P. Minetto, F. Paoloni, F. Cotugno* et al.
Journal of Clinical Oncology, Volume 41, Number 16 suppl. DOI: https://doi.org/10.1200/JCO.2023.41.16_suppl.7013

2023 **Pseudo-observations in survival analysis**, *M. Cipriani, A. Piciocchi, V. Arena, M. Alfò*.
Book of short papers SIS 2023 - Statistical Learning, Sustainability and Impact Evaluation. ISBN: 9788891935618

2022 **The Addition of Venetoclax to Induction Chemotherapy in No Low-Risk AML Patients: A Propensity Score-Matched Analysis of the Gimema AML1718 and AML1310 Trials**, *A. Piciocchi, M. Messina, M. Cipriani, F. Paoloni* et al.
Blood, vol. 140 (Supplement 1): 137–138. DOI: https://doi.org/10.1182/blood-2022-158890

2022 **Health-Related Quality of Life and Financial Burden of Ethiopian Patients with Chronic Myeloid Leukemia Receiving Tyrosine Kinase Inhibitors**, *F. Tadesse, A. Gebremedhin, A. Abubeker, A. Piciocchi, M. Cipriani* et al.
Blood, vol. 140 (Supplement 1): 5264–5265. DOI: https://doi.org/10.1182/blood-2022-163049

Preprint and under review

2026+ **Dynamic Prediction in Mixture Cure Models: A Model-Based Landmarking Approach**, *M. Cipriani, M. Alfò, M. Signorelli*.
arXiv preprint *arXiv:2509.18875 [stat.ME]*. DOI: https://doi.org/10.48550/arXiv.2509.18875

2026+ **A multiple imputation approach to distinguish curative from life-prolonging effects in the presence of missing covariates**, *M. Cipriani, M. Fiocco, M. Alfò, M. Quelhas, E. Musta*.
arXiv preprint *arXiv:2408.16485 [stat.ME]*. DOI: https://doi.org/10.48550/arXiv.2408.16485

Ph.D. Thesis

2026 **Flexible methods for survival data: synthetic patients, imputation and prediction in mixture cure models, M. Cipriani**, ***Supervisor: Prof. Marco Alfo', Prof. Marta Fiocco***.
Iris: https://iris.uniroma1.it/handle/11573/1760888